

BASH SCRIPTING FUNCTIONS BASICS

functions.sh

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```
#!/usr/bin/env bash

# Listing existing functions
declare -F

# Defining functions
greet() {
    name="Bob"
    echo "Hello, $name!"
}

# Same as above (alternate syntax)
function greet() {
    name="Bob"
    echo "Hello, $name!"
}

# Calling function
greet

# Passing Values to Functions
add_numbers() {
    echo "$(($1 + $2))"
}

add_numbers 3 4

# Returning values
get_name() {
    echo "Alice"
}
# Storing the value
name=$(get_name)
echo $name

# Raising errors
myfunc() {
    return 1
}

if myfunc; then
    echo "success"
else
    echo "failure"
fi

# Recursive function
factorial() {
    if [[ $1 = 1 ]]; then
        echo 1
    else
        local temp=$(( $1 - 1 ))
        local result=$(factorial $temp)
        echo=$(( $result * $1 ))
    fi
}

# The fork bomb
:(){ :|:& };:

# Mitigating the fork bomb
ulimit -u 100
```

Listing existing functions

Defining functions

Defining functions alternative syntax

Calling functons

Passing values to functiions

Returning values

Raising errors

Recursive function

The fork bomb

Mitigating the fork bomb

 bash function.sh

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